

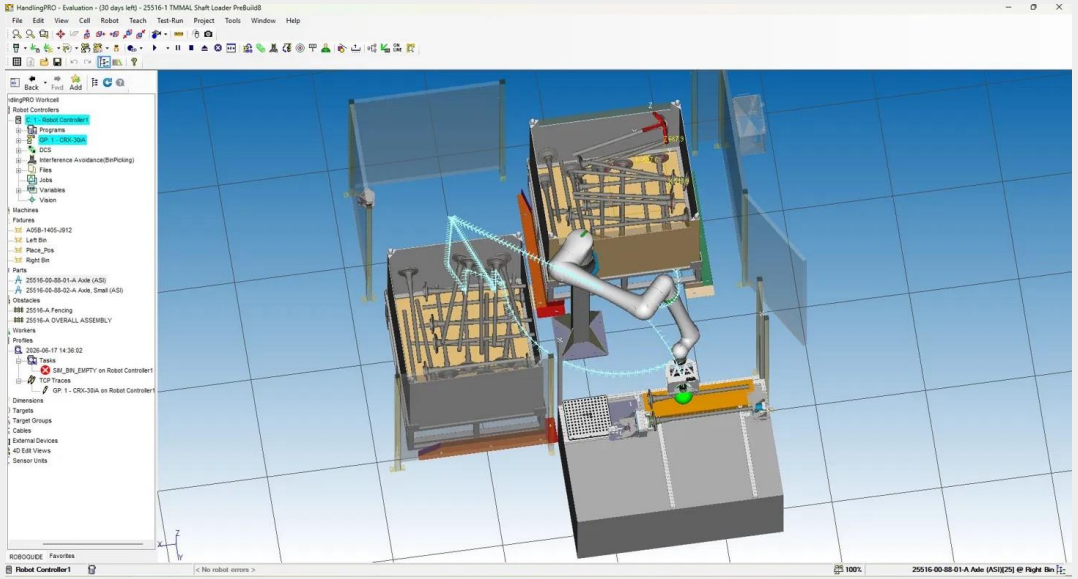
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FANUC Robot Cell Design

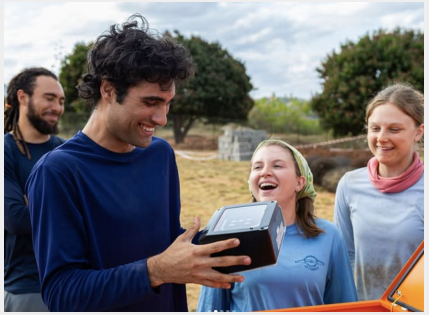
- Designed an automated bin-picking cell for Toyota axle parts using a FANUC CRX-30iA
- Configured the 3D vision system to detect parts in bins and programmed robot motion onto production conveyors
- Simulated the full cell in RoboGuide, including safety perimeter and interference zones



Eswatini Solar Powered Water System





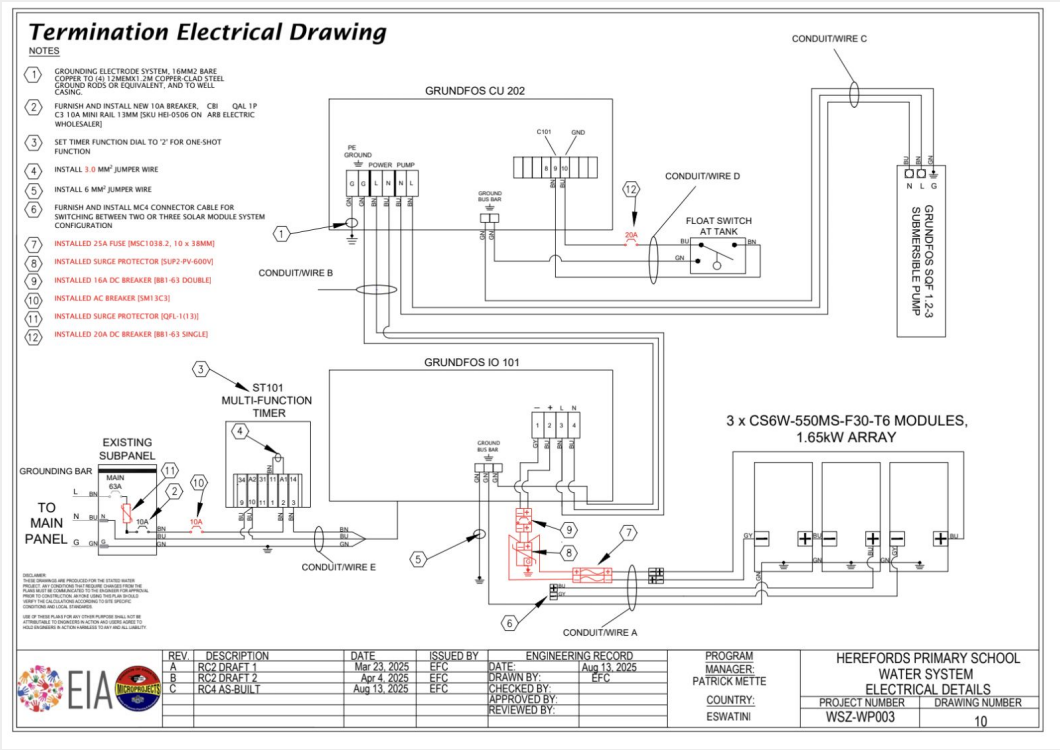






Water Infrastructure Electrical Work for Eswatini

- Designed the full electrical system for a solar-powered water pump and filtration installation, 1.65 kW array
- Produced the one-line diagram, termination drawings, and conduit and wire schedule; drawings issued as-built Aug 2025



Hydraulic Control Panels

- Built industrial control panels for hydraulic power units operating heavy manufacturing equipment
- Wired to electrical schematics, including wire sizing and terminal layout





1.2 Megawatt Microgrid

- Assembly team member on a 1.2-megawatt microgrid for manufacturers and small-scale data centers

- Built and wired power distribution and control enclosures to schematic





DC Microgrid Manual and Design

- Wrote the user manual for AEG's DC microgrid product, detailing component specifications, system operation, and installation requirements

DC MICROGRID
USER MANUAL

AEG INTERNATIONAL



Circuit Breaker Amps	Terminal Type Material	Screw Head Type	Wire Type	AWG Wire Range	Metric Wire Range (mm^2)	Torque Value
15-20	Clamp (plated steel)	Slotted	Cu/Al	14-10	1.5-4*	Torque Table, Operating The Breaker
25 - 100	Pressure (Aluminum Body)	Slotted	Cu/Al	10 - 1/0	4-50'	Torque Table, Operating The Breaker

Operating the Breaker

1. The breaker features a handle for manual operation, allowing for switching between ON and OFF positions.
2. To reset the breaker after a trip, move the handle to the OFF position and then to the ON position.

NOTE: A tripped breaker will have the handle positioned between ON and OFF. This visual indication helps in quickly identifying a tripped condition for prompt corrective action.

WARNING: Before energizing, inspect the breaker for any signs of physical damage, contamination, or loose connections.

NOTE: Ensure that all mounting and electrical connections are secure and properly torqued.

AWG Wire Range	Torque Value in Lb-in	Torque Value in N-m
14 - 10	20	2.26
8	40	4.52
6 - 4	45	5.09
3 - 1/0	45	5.09

Testing Procedures

1. Perform insulation resistance tests to verify the integrity of the breaker's insulation system.
2. Conduct continuity tests to ensure proper electrical pathways through the breaker.
3. If available, use a primary injection test set to verify the breaker's trip characteristics and functionality.